Problem Solving Review

Worked answer key: question, given, diagram, steps with reasons, final answer

Conventions used throughout

• **Notation:** v_1 = initial velocity, v_2 = final velocity, Δd = displacement, Δt = time interval, a = acceleration, $\vec{a}_{av}$ = average acceleration, v_{ave} = average speed. An arrow marks a vector; bars, as in $|\Delta \vec{v}|$, mean its size only. In relative-velocity problems v_{AB} means the velocity of A relative to B.

• **Which object?** When two objects move in one problem, a word in the subscript says which one a symbol belongs to. For example, a_{car} means the car's acceleration.

• **Constant-acceleration equations** (valid only when a is constant): $v_2 = v_1 + a\Delta t$; $\Delta d = \frac{1}{2}(v_1 + v_2)\Delta t$; $\Delta d = v_1\Delta t + \frac{1}{2}a\Delta t^2$; $\Delta d = v_2\Delta t - \frac{1}{2}a\Delta t^2$; $v_2^2 = v_1^2 + 2a\Delta d$.

• **Signs:** each question states which direction is positive. A negative value means the opposite direction. Final answers state the real direction.

• **$g = 9.81\ m/s^2$** and no air resistance. Full precision is kept in every step; only final answers are rounded.

REVIEW 1

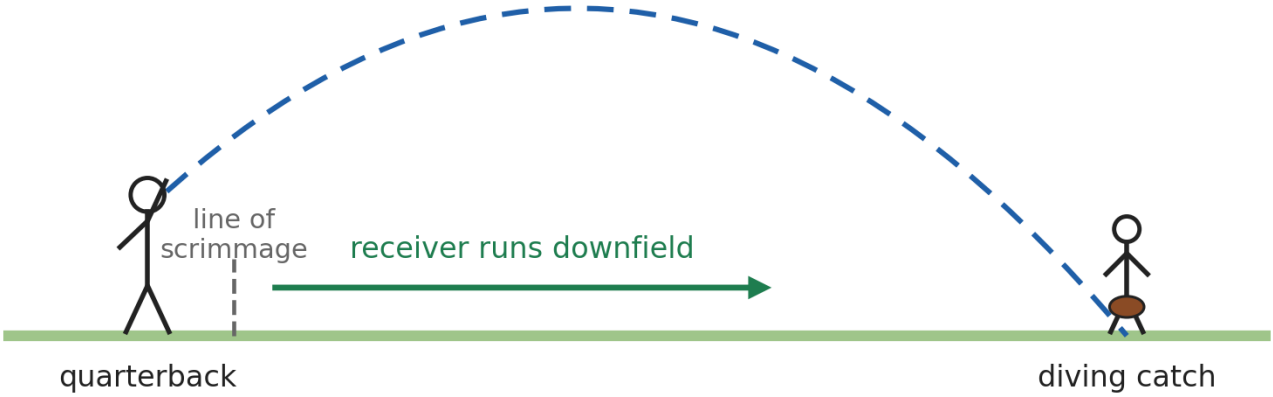
Football pass and receiver

A football quarterback attempts a pass to one of the receivers. As the ball is snapped, the receiver leaves the line of scrimmage and runs directly down field. The quarterback releases the ball 2.0 s later and from a position 3.0 m behind the line of scrimmage. He throws the ball with a speed of 26 m/s at an elevation of 60° above the horizontal. The receiver makes a diving reception, catching the ball just as it reaches the ground. See the diagram below.

a. What is the time of flight of the football?

b. What is the average speed of the receiver?

Picture the situation



At the snap the receiver starts running downfield. Two seconds later the quarterback, standing 3.0 m behind the line, throws a high pass. The receiver dives and catches it just as it reaches the ground.

SOLVING FOR

(a) how long the ball is in the air, Δt; (b) the receiver's average speed from the snap to the catch.

GIVEN

$v_1 = 26$ m/s at 60°	$\Delta d_y = -2.0$ m	$a_y = -9.81$ m/s^2
$x_{release} = -3.0$ m	delay $= 2.0$ s	$\Delta t = ?$

Sign convention: +x is downfield and +y is up. The line of scrimmage is x = 0. The ball ends 2.0 m lower than it starts (released 2.0 m up, caught at the ground), so $\Delta d_y = -2.0$ m.

Diagram: the play (drawn to scale)

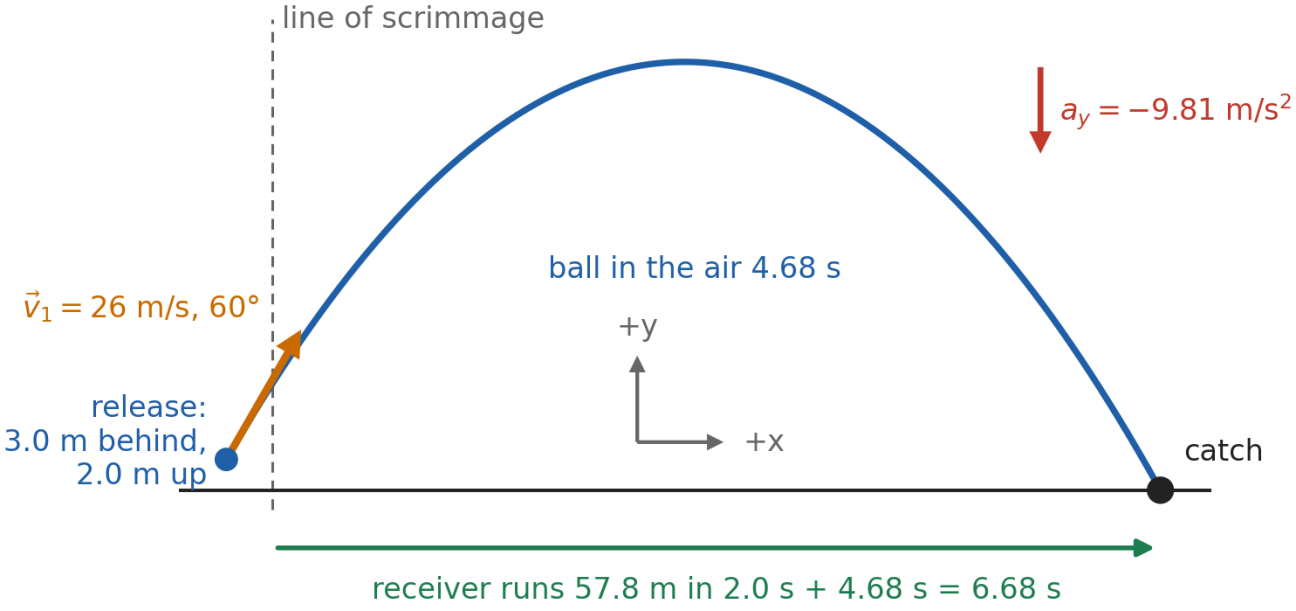
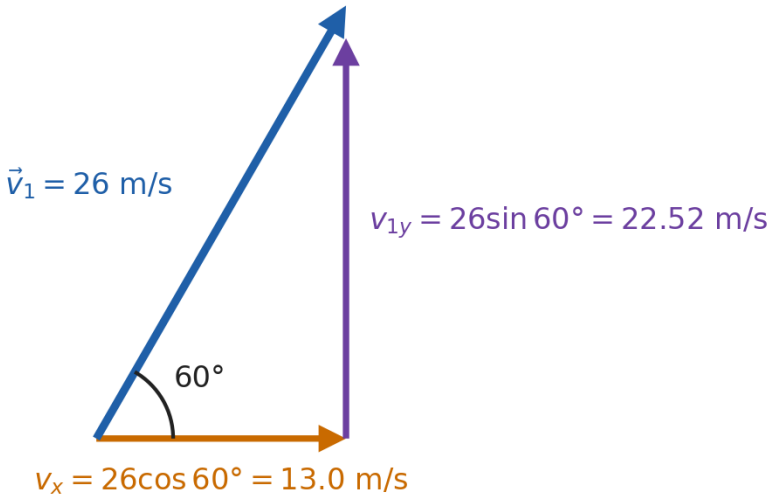


Diagram: splitting the launch velocity into parts



Step 1 – (a) Time in the air, from the vertical motion

This is a projectile, so we treat up-down and sideways separately. The flight ends when the ball reaches the ground, which is a fact about its height, so the vertical motion gives the time. Vertically we know the up-down part of the launch velocity, v_{1y}, plus a_y and Δd_y. Putting them into the displacement equation gives a quadratic in Δt.

Formula: $v_{1y} = v_1 \sin\theta = 26 \sin 60° = 22.52$ m/s

Formula: $\Delta d_y = v_{1y}\Delta t + \frac{1}{2} a_y \Delta t^2$

Put in the numbers:

$$-2.0 = 22.52\,\Delta t + \frac{1}{2}(-9.81)\,\Delta t^2 = 22.52\,\Delta t - 4.905\,\Delta t^2$$

Move everything to the left side:

$$4.905\,\Delta t^2 - 22.52\,\Delta t - 2.0 = 0$$

Formula: $\Delta t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Here $a = 4.905,\ b = -22.52,\ c = -2.0$:

$$\Delta t = \frac{22.52 \pm \sqrt{(-22.52)^2 - 4(4.905)(-2.0)}}{2(4.905)}$$

$$\Delta t = \frac{22.52 \pm \sqrt{507.2 + 39.2}}{9.81}$$

$$\Delta t = \frac{22.52 \pm 23.37}{9.81} \Rightarrow \Delta t = \frac{45.89}{9.81} = 4.68 \text{ s} \quad \text{or} \quad \frac{-0.85}{9.81} = -0.09 \text{ s}$$

The "−" choice gives a negative time (before the throw), so we reject it. **Δt = 4.68 s**.

Step 2 – Where the ball lands

Sideways the ball moves at a steady $v_x = 26 \cos 60° = 13.0$ m/s for those 4.68 s. That tells us how far it travels from the quarterback. The quarterback was 3.0 m behind the line, so we subtract 3.0 m to get the distance from the line, which is where the receiver started.

Formula: $v_x = v_1 \cos\theta = 26 \cos 60° = 13.0 \text{ m/s}$

Formula: $\Delta d_x = v_x \Delta t$

$\Delta d_x = (13.0 \text{ m/s})(4.678 \text{ s}) = 60.81 \text{ m}$

$\text{catch point} = 60.81 - 3.0 = 57.81 \text{ m past the line}$

Step 3 – (b) The receiver's average speed

The receiver ran from the line (at the snap) to the catch point. He ran during the 2.0 s before the throw *and* the 4.68 s the ball was in the air. $\Delta t_{receiver}$ is his total running time and d is the distance he ran. Average speed = total distance ÷ total time.

$\Delta t_{receiver} = 2.0 + 4.678 = 6.678 \text{ s}$

Formula: $v_{ave} = \dfrac{d}{\Delta t}$

$v_{ave} = \dfrac{d}{\Delta t_{receiver}} = \dfrac{57.81 \text{ m}}{6.678 \text{ s}} = 8.66 \text{ m/s}$

(a) 4.7 s

(b) 8.7 m/s

REVIEW 2

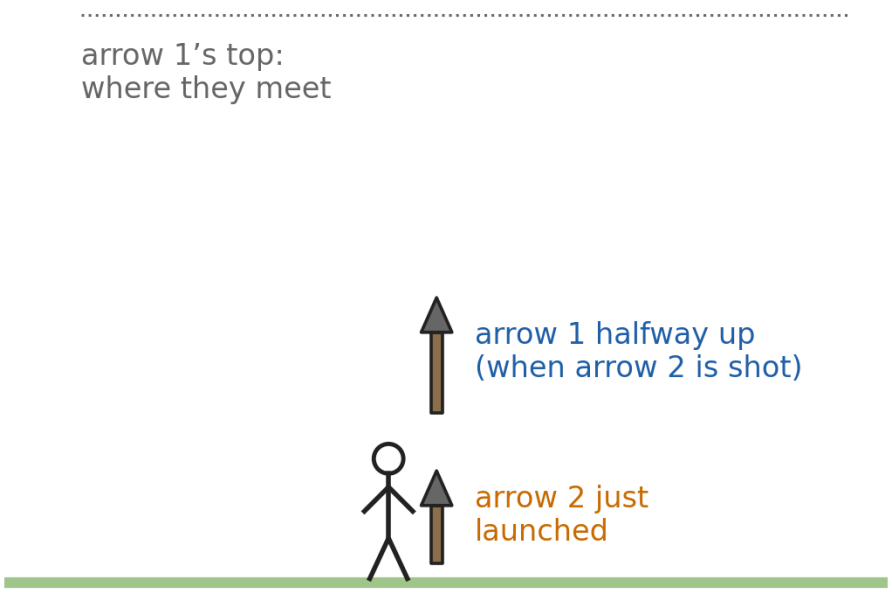
Two vertical arrows

An arrow is shot vertically upward with an initial speed of 25 m/s. When it's exactly halfway to the top of its flight, a second arrow is launched vertically upward from the same spot. The second arrow reaches the first arrow just as the first arrow reaches its highest point.

a. What is the launch speed of the second arrow?

b. What maximum height does the second arrow reach?

Picture the situation



The first arrow is already halfway up when the second one is fired from the same spot. The second arrow is faster and catches up with the first exactly at the first arrow's highest point.

GIVEN

$v_{1(\text{arrow 1})} = +25$ m/s

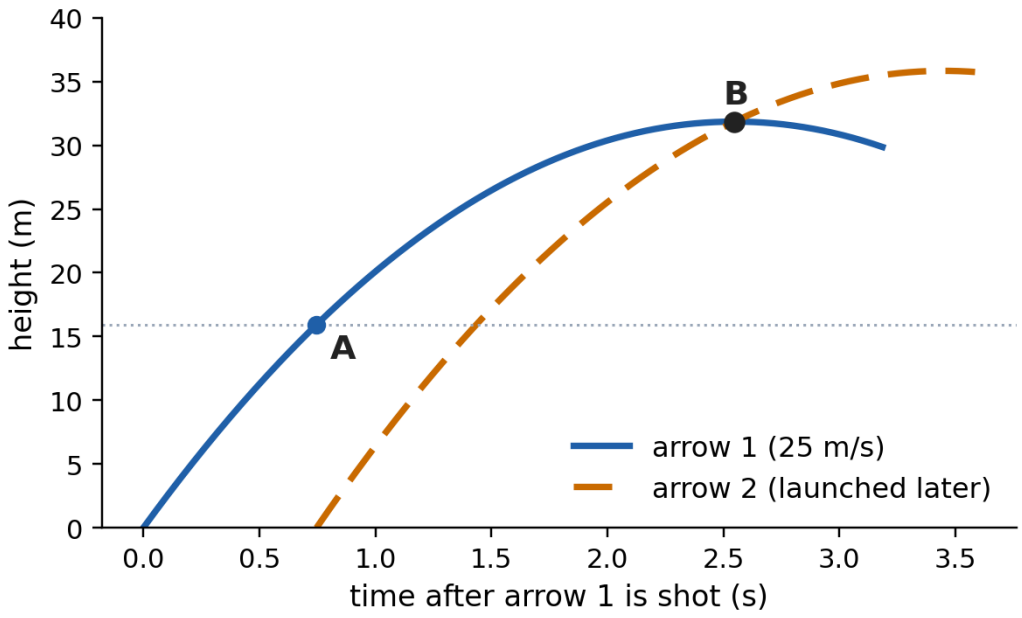
$a = -9.81$ m/s^2

$v_{1(\text{arrow 2})} = ?$

$H = ?$ (height of arrow 1's top)

Sign convention: up is positive, so a = −9.81 m/s^2 for both arrows. Each arrow has its own launch speed, so the label in brackets says which arrow a v_1 belongs to: $v_{1(\text{arrow 1})}$ is arrow 1's launch speed and $v_{1(\text{arrow 2})}$ is arrow 2's.

Graph: height vs time for both arrows (A: arrow 2 fired when arrow 1 is halfway up; B: they meet at the top)



Step 1 – Where and when arrow 1 reaches its top

Everything in this problem is measured against arrow 1's top, so find that first. At the top an arrow stops for an instant, so $v_2 = 0$. Use $v_2^2 = v_1^2 + 2a\Delta d$ for the height and $v_2 = v_1 + a\Delta t$ for the time. Here $v_1 = 25$ m/s (arrow 1's launch speed), and the displacement Δd up to the top is the height H we want. Δt_{top} is the time arrow 1 takes to reach its top.

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_2 = 0$ and $\Delta d = H$: $0 = 25^2 + 2(-9.81)H = 625 - 19.62H$

Add $19.62H$ to both sides, then divide by 19.62:

$$H = \frac{625}{19.62} = 31.86 \text{ m}$$

Formula: $v_2 = v_1 + a\Delta t$

Put in $v_2 = 0$: $0 = 25 + (-9.81)\Delta t_{top}$

Add $9.81\Delta t_{top}$ to both sides, then divide by 9.81:

$$\Delta t_{top} = \frac{25}{9.81} = 2.548 \text{ s}$$

Step 2 – When arrow 1 is halfway up

Arrow 2 is fired when arrow 1 is at H/2 = 15.93 m (on the way up). We need that moment. First find arrow 1's speed at that height (no time needed), then use the speed to get the time. Here v_2 is arrow 1's speed at the halfway point, and Δt_{half} is the time arrow 1 takes to get there.

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

$v_2^2 = 25^2 + 2(-9.81)(15.93) = 625 - 312.5 = 312.5$

Square root both sides: $v_2 = \sqrt{312.5} = 17.68$ m/s

Formula: $v_2 = v_1 + a\Delta t$

$17.68 = 25 + (-9.81)\Delta t_{half}$

Subtract 25 from both sides: $-7.32 = -9.81\,\Delta t_{half}$

Divide both sides by -9.81: $\Delta t_{half} = \dfrac{-7.32}{-9.81} = 0.746$ s

Step 3 – (a) Launch speed of arrow 2

Arrow 2 is fired at 0.746 s and must be at 31.86 m when arrow 1 gets there, at 2.548 s. So arrow 2 flies for 2.548 − 0.746 = 1.802 s and rises 31.86 m in that time. In $\Delta d = v_1\Delta t + \frac{1}{2}a\Delta t^2$, v_1 is the starting velocity of whatever is moving. The thing moving here is arrow 2, so its launch speed $v_{1(\text{arrow 2})}$ goes in the v_1 spot. It is the only unknown:

Formula: $\Delta d = v_1\Delta t + \dfrac{1}{2}a\Delta t^2$

$31.86 = v_{1(\text{arrow 2})}(1.802) + \dfrac{1}{2}(-9.81)(1.802)^2$

Work out the last term: $31.86 = 1.802\,v_{1(\text{arrow 2})} - 15.93$

Add 15.93 to both sides: $47.79 = 1.802\,v_{1(\text{arrow 2})}$

Divide both sides by 1.802: $v_{1(\text{arrow 2})} = \dfrac{47.79}{1.802} = 26.52$ m/s

Step 4 – (b) How high arrow 2 goes

When they meet, arrow 2 is still moving up (it was launched faster), so it keeps rising after passing arrow 1. Its own top is where its velocity reaches zero. This time the motion is arrow 2's, so v_1 is $v_{1(\text{arrow 2})}$ = 26.52 m/s, and Δd is arrow 2's maximum height, $H_{\text{arrow 2}}$:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_2 = 0$ and $\Delta d = H_{\text{arrow 2}}$:

$$0 = (26.517)^2 + 2(-9.81)H_{\text{arrow 2}} = 703.1 - 19.62H_{\text{arrow 2}}$$

Add $19.62H_{\text{arrow 2}}$, then divide by 19.62:

$$H_{\text{arrow 2}} = \frac{703.1}{19.62} = 35.84 \text{ m}$$

(a) 27 m/s [up]

(b) 36 m above the launch point

Check: arrow 2's speed when they meet is 26.52 − 9.81(1.802) = 8.84 m/s [up]. It is still positive, so arrow 2 really is still rising and does go higher than 31.9 m.

REVIEW 3

Truck and accelerating car

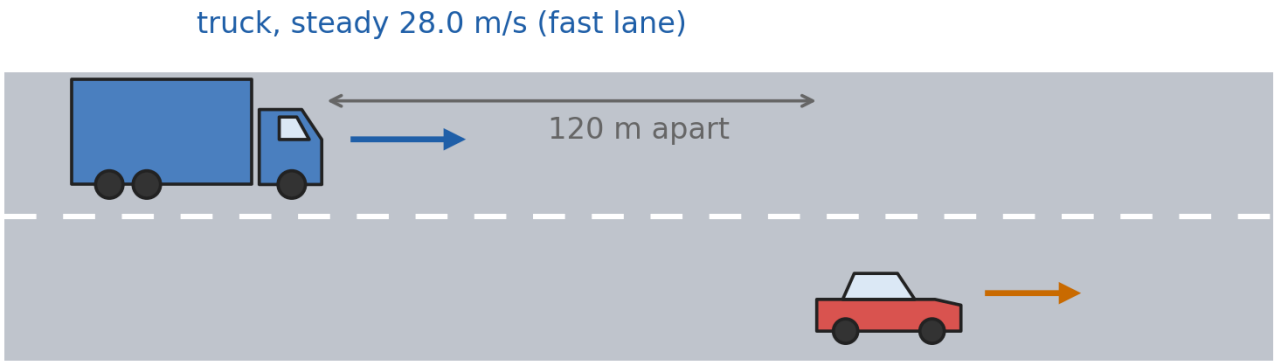
A truck travels at a constant speed of 28.0 m/s in the fast lane of a two-lane highway. It approaches a stationary car stopped at the side of the road. When the truck is still 1.2 x 10^2 m behind the car, the car pulls out into the slow lane with an acceleration of 2.6 m/s^2.

a. How long will it take the truck to pass the car?

b. How far will the car have travelled when the truck passes it?

c. If the car were to maintain this acceleration, what would be its speed when it overtakes the truck?

Picture the situation



The truck comes up from behind at a steady speed. The car starts from rest and speeds up. At first the truck is faster and passes the car; later the car is faster and passes the truck back.

GIVEN

$v_{truck} = 28.0$ m/s $\qquad a_{car} = 2.6$ m/s^2 $\qquad$ gap $= 1.2 \times 10^2$ m

Sign convention: + is the direction of travel. All positions are measured from where the car starts (x = 0), so the truck starts at x = −120 m.

Step 1 – Write each vehicle's position

"Passing" means both are at the same position at the same time. So we write each one's position as a formula in Δt, starting from its own starting position. In these formulas x is the position at time Δt and x_1 is the starting position. The truck moves at constant velocity; the car starts from rest with constant acceleration.

Formula (truck, constant velocity): $x = x_1 + v\,\Delta t$

$$x_{truck} = -120 + 28.0\,\Delta t$$

Formula (car): $x = x_1 + v_1\Delta t + \frac{1}{2}a\Delta t^2$

$$x_{car} = 0 + (0)\Delta t + \frac{1}{2}(2.6)\,\Delta t^2 = 1.3\,\Delta t^2$$

Step 2 – Set the positions equal

Setting $x_{car} = x_{truck}$ and moving everything to one side gives a quadratic.

$$1.3\,\Delta t^2 = -120 + 28.0\,\Delta t$$

Move everything to the left (add 120, subtract $28.0\Delta t$):

$$1.3\,\Delta t^2 - 28.0\,\Delta t + 120 = 0$$

Formula: $\Delta t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

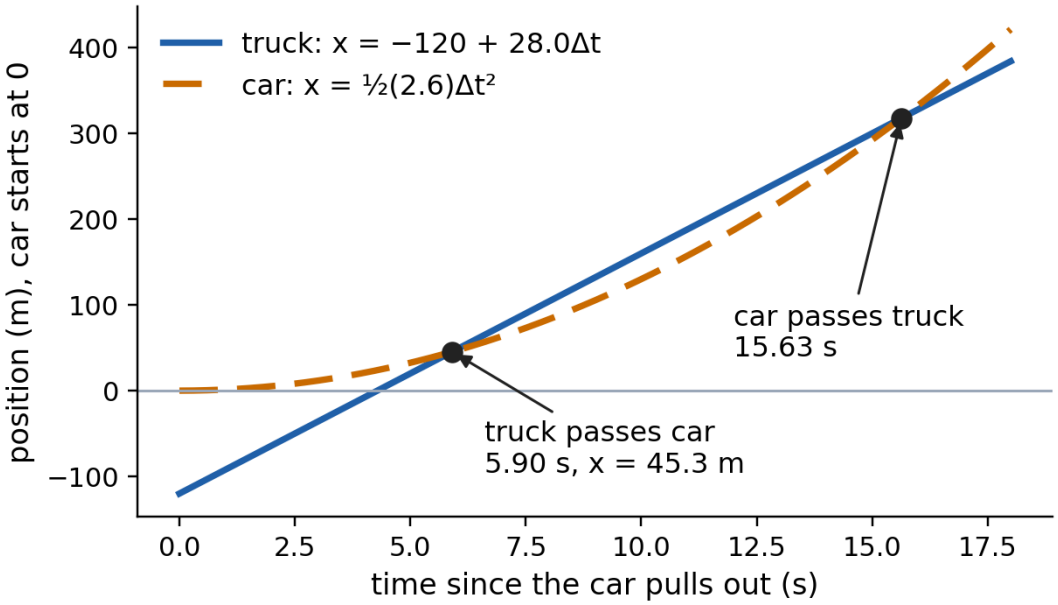
Here $a = 1.3,\ b = -28.0,\ c = 120$:

$$\Delta t = \frac{28.0 \pm \sqrt{784 - 624}}{2.6} = \frac{28.0 \pm \sqrt{160}}{2.6} = \frac{28.0 \pm 12.65}{2.6}$$

$$\Delta t = 5.90 \text{ s} \quad \text{or} \quad \Delta t = 15.63 \text{ s}$$

Both answers are real meetings. At 5.90 s the truck catches and passes the car. The car keeps speeding up, so at 15.63 s it catches and passes the truck. Part (a) asks about the first; part (c) about the second.

Graph: positions of the truck and the car



Step 3 – (b) How far the car has gone when the truck passes

Put the first time into the car's position formula (the car started at x = 0, so its position is its distance travelled).

$$x_{car} = 1.3\,\Delta t^2 = 1.3(5.904)^2 = 1.3(34.86) = 45.3 \text{ m}$$

Step 4 – (c) The car's speed when it overtakes the truck

Use the later time in $v_2 = v_1 + a\Delta t$ for the car: $v_1 = 0$ because the car started from rest, $a = a_{car} = 2.6$ m/s², and v_2 is the car's speed when it overtakes.

Formula: $v_2 = v_1 + a\Delta t$

$$v_2 = 0 + (2.6)(15.63) = 40.6 \text{ m/s}$$

(a) 5.9 s

(b) 45 m

(c) 41 m/s

Check: at 5.904 s the truck is at −120 + 28.0(5.904) = 45.3 m, the same place as the car.

REVIEW 4

Clock hand: speed and changing velocity

A clock has a second hand 15 cm long, from centre to tip. (Hint: the velocity of the tip of the second hand is always tangent to the circle)

a. Draw a labelled diagram to represent this situation

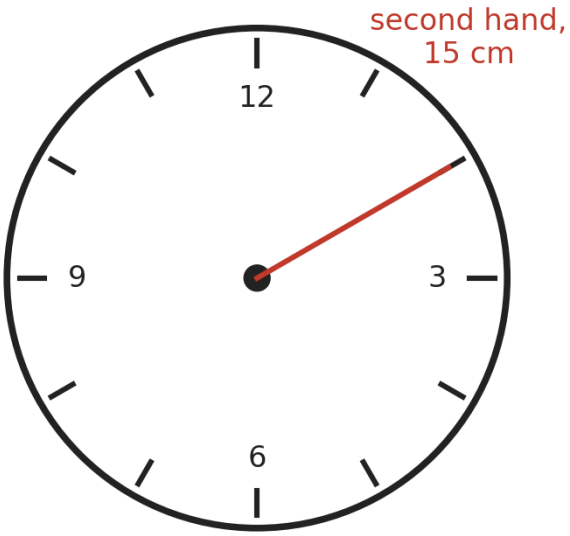
b. Calculate the instantaneous speed of the tip of the second hand.

c. What is the instantaneous velocity of the tip at 15 s?

d. What is its change in velocity between 30 s and 45 s?

e. What is its average acceleration during the same time interval? (1 mark)

Picture the situation



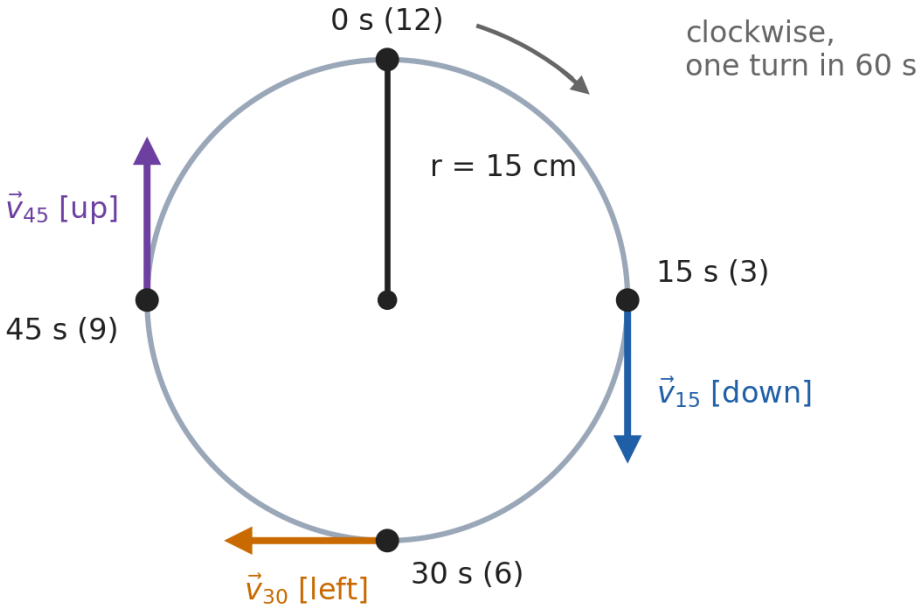
The tip of the second hand goes around a circle of radius 15 cm once every minute. Its speed never changes, but its direction keeps turning.

GIVEN

$r = 15$ cm $\qquad T = 60$ s per turn $\qquad \Delta t = 45 - 30 = 15$ s

Sign convention: up and right are positive. The hand turns clockwise and points at 12 at 0 s. r is the length of the hand (the radius of the circle) and T is the time for one full turn. A number in a velocity's subscript is the clock time: $\vec{v}_{15}$, $\vec{v}_{30}$ and $\vec{v}_{45}$ are the tip's velocity at 15 s, 30 s and 45 s.

Diagram (a): the clock face with velocities at 15 s, 30 s and 45 s



(b) Speed of the tip

In one full turn the tip travels once around the circle, a distance of 2πr (the circumference), and that takes one period, T = 60 s. Speed = distance ÷ time.

Formula: $v = \frac{d}{T}$ with $d = 2\pi r$ (circumference), $T = 60$ s

$$v = \frac{2\pi(15\text{ cm})}{60\text{ s}} = \frac{94.25\text{ cm}}{60\text{ s}} = 1.57\text{ cm/s}$$

(c) Velocity at 15 s

After 15 s the hand points at the 3. The hint says the velocity is tangent to the circle, along the direction of motion. Going clockwise from the 3, the tip is moving straight down.

$\vec{v}_{15}$ = 1.57 cm/s [down].

(d) Change in velocity from 30 s to 45 s

At 30 s the tip is at the 6 and moving left. At 45 s it is at the 9 and moving up. Same speed, different directions. $\Delta\vec{v} = \vec{v}_{45} + (-\vec{v}_{30})$, and reversing the leftward $\vec{v}_{30}$ gives an arrow pointing right. Up and right are perpendicular and equal in length, so Pythagoras gives the size and the angle is 45°.

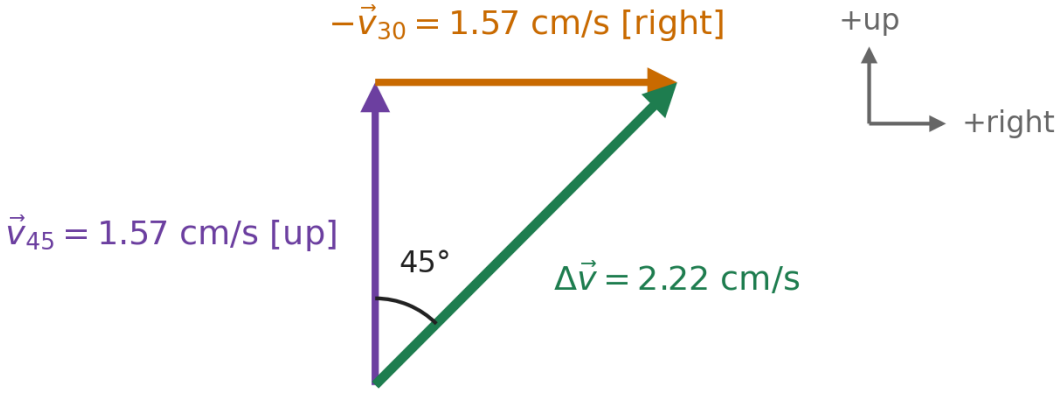
Formula: $\Delta\vec{v} = \vec{v}_{45} - \vec{v}_{30} = \vec{v}_{45} + (-\vec{v}_{30})$

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$$|\Delta\vec{v}| = \sqrt{(1.571)^2 + (1.571)^2} = \sqrt{4.936} = 2.22 \text{ cm/s}$$

The direction is 45° between up and right, which points toward the centre of the clock from where the tip is halfway through this interval (between the 6 and the 9).

Diagram (d): $\vec{v}_{45} + (-\vec{v}_{30})$, tip to tail



(e) Average acceleration

Divide the change in velocity by the 15 s it took. The direction stays the same. $\vec{a}_{av}$ is the average acceleration, and $|\Delta\vec{v}|$ (with bars) means the size of $\Delta\vec{v}$ only.

Formula: $\vec{a}_{av} = \dfrac{\Delta\vec{v}}{\Delta t}$

$\vec{a}_{av} = \dfrac{2.22 \text{ cm/s}}{15 \text{ s}} = 0.148 \text{ cm/s}^2$, same direction as $\Delta\vec{v}$

(b) 1.6 cm/s

(c) 1.6 cm/s [down]

(d) 2.2 cm/s [up 45° right] (up and to the right at 45°)

(e) 0.15 cm/s^2 [up 45° right]

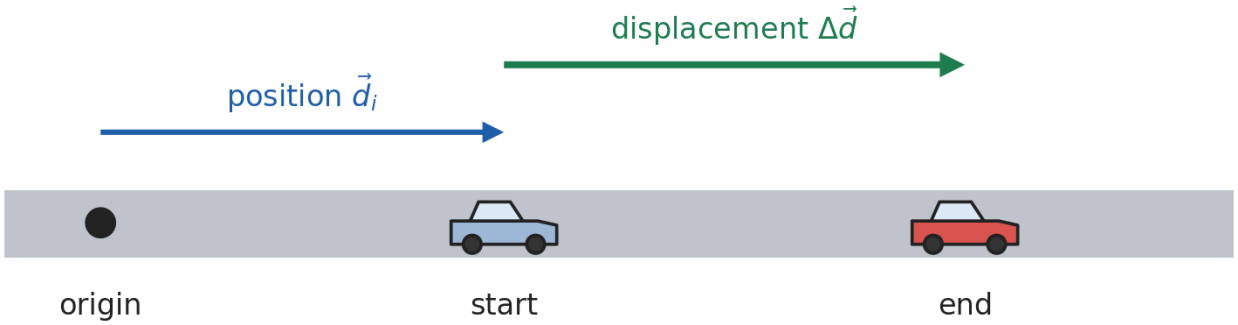
The acceleration points toward the inside of the circle. Any object moving in a circle at constant speed accelerates toward the centre, because its velocity is always being turned toward the centre.

CONCEPTS 1–2

Counts, velocity and displacement notation

1. How many people are there in this room? _____ How many significant digits is this? _____
2. Explain the difference between the following symbols:
   a. $\vec{v}_{av}$, $\Delta\vec{v}$, $\vec{v}_i$
   b. $\vec{d}$, $\Delta\vec{d}$

Picture the situation



Position is measured from a chosen origin. Displacement is the change in position: the arrow from where something started to where it ended.

1. Counting people

I do not know how many people were in your room, so count them. The key idea is that a count of whole objects is **exact**: there is no measuring error in "24 people". Exact numbers have unlimited (infinitely many) significant digits, so they never limit the significant figures of an answer.

2(a) $\vec{v}_{av}$, $\Delta\vec{v}$ and $\vec{v}_i$

All three have units of m/s, but they mean different things.

$$\vec{v}_{av} = \frac{\Delta\vec{d}}{\Delta t} \qquad \Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

$\vec{\mathbf{v}}_{\mathbf{av}}$, average velocity: the displacement divided by the time taken (one velocity that describes the whole trip). $\boldsymbol{\Delta\vec{v}}$, change in velocity: how much the velocity changed, final minus initial. $\vec{\mathbf{v}}_{\mathbf{i}}$, initial velocity: the velocity at the start of the time interval ($\vec{v}_f$ is the velocity at the end).

2(b) $\vec{d}$ and $\Delta\vec{d}$

$$\Delta\vec{d} = \vec{d}_f - \vec{d}_i$$

$\vec{d}$, position: where an object is, measured from a chosen origin (as in the sketch). **$\Delta\vec{d}$**, displacement: the change in position, from the starting point to the ending point ($\vec{d}_i$ = initial position, $\vec{d}_f$ = final position). If you move the origin, every position changes, but the displacement between two points does not.

1. Your own head count; it is exact, so unlimited significant digits

2(a) average velocity; change in velocity; initial velocity

2(b) position (from an origin); displacement (change in position)

CONCEPT 3

Toddlers with different starting times

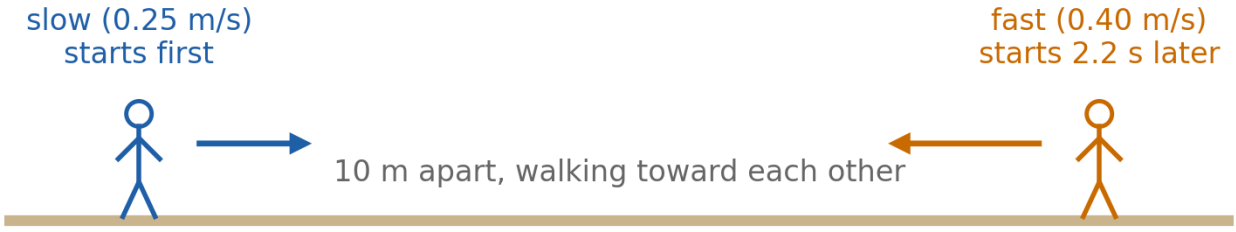
Two toddlers are 10 m apart and are moving towards each other, one travelling at 0.4 m/s and the other at 0.25 m/s. The faster child started 2.2 s after the slower one. Which of the following statements are true at the time the two children meet? (s = slow; f = fast)

a. $\Delta t_f = \Delta t_s + 2.2s$

b. $\Delta t_s = \Delta t_f + 2.2s$.

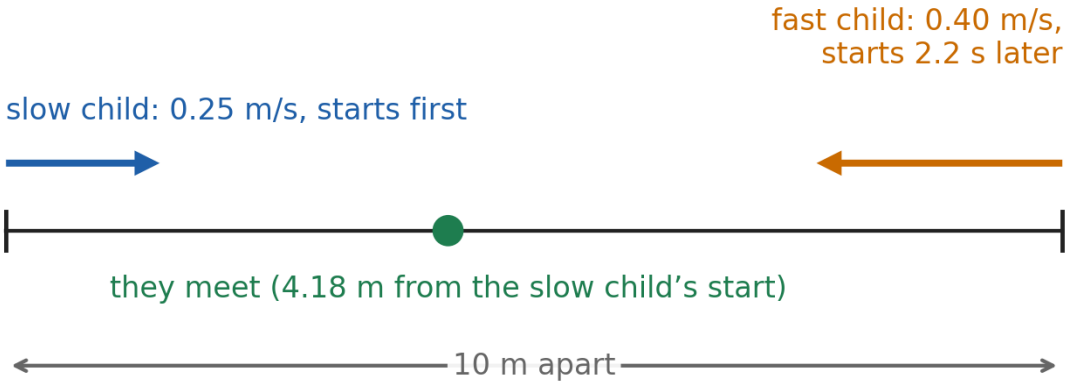
Explain your answer:

Picture the situation



Two toddlers walk toward each other. The slow one sets off first; 2.2 s later the fast one sets off.

Diagram: the situation (drawn to scale)



Explanation

Think about each toddler’s stopwatch. The slow toddler starts walking first. By the time the fast toddler starts, the slow toddler’s stopwatch already reads 2.2 s. They both stop when they meet. So the slow toddler has always been walking 2.2 s longer than the fast one.

$$\Delta t_s = \Delta t_f + 2.2 \text{ s}$$

Statement (a) says the opposite (the fast toddler walks longer), so it is false.

Optional: when and where they meet

The two walking distances must add up to the 10 m gap. d_s and d_f are the distances walked by the slow and fast toddler. Writing the slow toddler's time as $\Delta t_f + 2.2$ leaves one unknown.

Formula: $d = v\,\Delta t$

The distances fill the gap: $d_s + d_f = 10$

Put in $d_s = 0.25\,\Delta t_s = 0.25(\Delta t_f + 2.2)$ and $d_f = 0.40\,\Delta t_f$:

$0.25(\Delta t_f + 2.2) + 0.40\,\Delta t_f = 10$

Expand the bracket: $0.25\,\Delta t_f + 0.55 + 0.40\,\Delta t_f = 10$

Add the like terms: $0.65\,\Delta t_f + 0.55 = 10$

Subtract 0.55, then divide by 0.65: $\Delta t_f = \dfrac{9.45}{0.65} = 14.54$ s

$\Delta t_s = 14.54 + 2.2 = 16.74$ s

The slow toddler walks 0.25 × 16.74 = 4.18 m and the fast one 0.40 × 14.54 = 5.82 m (total 10 m).

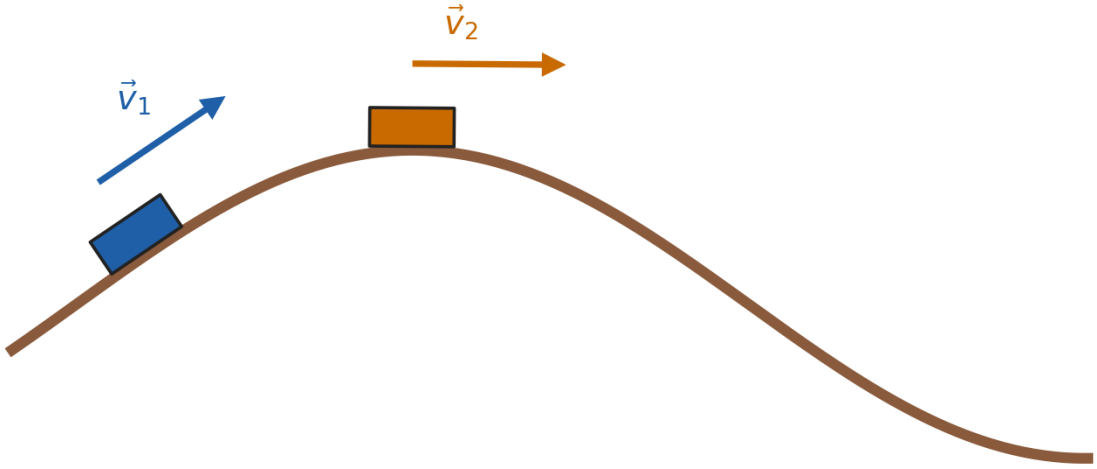
Only (b) is true: $\Delta t_s = \Delta t_f + 2.2$ s, because the slow child walked for 2.2 s longer.

CONCEPT 4

Roller coaster: acceleration direction

The following vectors show the velocity of a roller coaster at two different points in time. What is the direction of the acceleration? Explain.

Picture the situation



The coaster car climbs a hill (moving up and to the right), then levels off at the top (moving straight right).

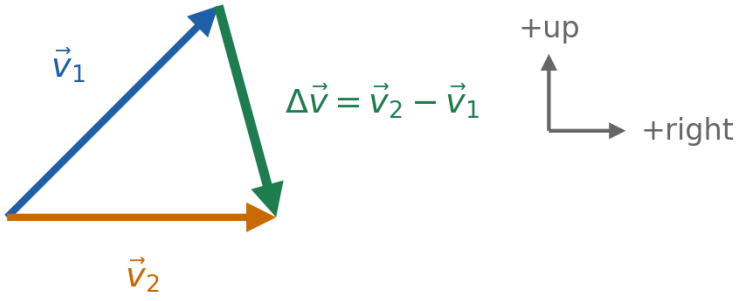
Step 1 – Acceleration points along the change in velocity

Average acceleration is the change in velocity divided by the time. Time is positive, so the acceleration points the same way as the change in velocity. So the question is really: which way did the velocity change?

$$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t}$$

Here $\vec{v}_1$ is the first velocity (climbing) and $\vec{v}_2$ is the later one (level at the top).

Diagram: tails together, $\Delta\vec{v}$ from the tip of $\vec{v}_1$ to the tip of $\vec{v}_2$



Step 2 – Read the direction off the diagram

Draw both velocities from the same starting point. The change $\Delta\vec{v}$ is the arrow from the tip of $\vec{v}_1$ to the tip of $\vec{v}_2$, since $\vec{v}_1 + \Delta\vec{v} = \vec{v}_2$. Here the upward part of the velocity disappears (a change downward) and the rightward part grows (a change to the right). So $\Delta\vec{v}$ points down and to the right.

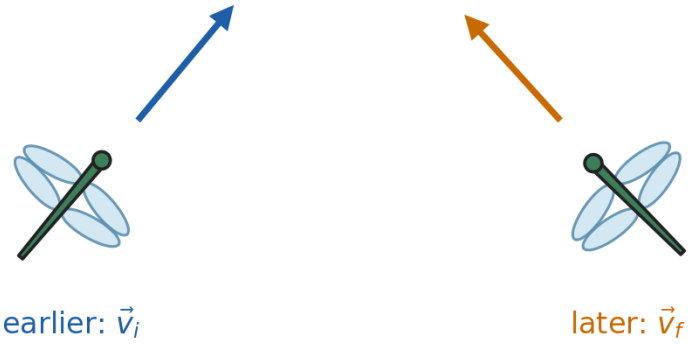
The acceleration points down and to the right, the direction of $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$.

CONCEPT 5

Dragonfly: subtract velocity vectors

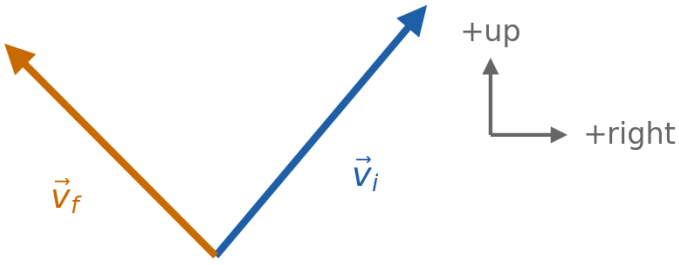
The velocity of a dragonfly at two different times is shown by the vectors below. Draw a vector diagram that would allow you to calculate the dragonfly's change in velocity.

Picture the situation



The dragonfly is first flying up and to the right, and later up and to the left.

The two velocities

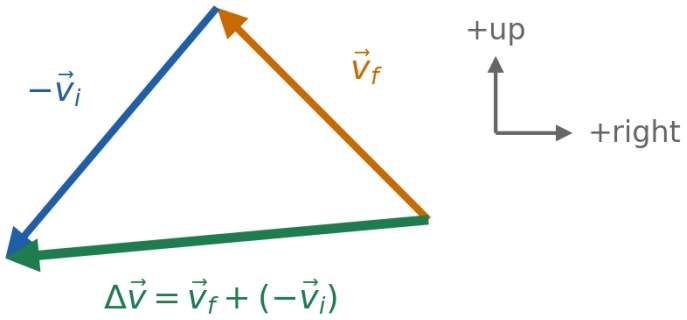


Step 1 – Turn the subtraction into addition

Change in velocity is final minus initial. We can only add arrows tip to tail, so we rewrite the subtraction as adding the opposite: reverse $\vec{v}_i$ (same length, opposite direction). Here $\vec{v}_i$ (i for initial) is the earlier velocity, up and to the right, and $\vec{v}_f$ (f for final) is the later one, up and to the left.

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i = \vec{v}_f + (-\vec{v}_i)$$

Vector diagram: $\vec{v}_f + (-\vec{v}_i)$, tip to tail



Step 2 – How to use the diagram

Draw $\vec{v}_f$. Put the tail of $-\vec{v}_i$ on its tip. The change in velocity, $\Delta\vec{v}$, is the arrow from the very first tail to the very last tip. To get numbers, measure $\Delta\vec{v}$ with a ruler and protractor using the worksheet's scale, or solve the triangle with the cosine law and sine law.

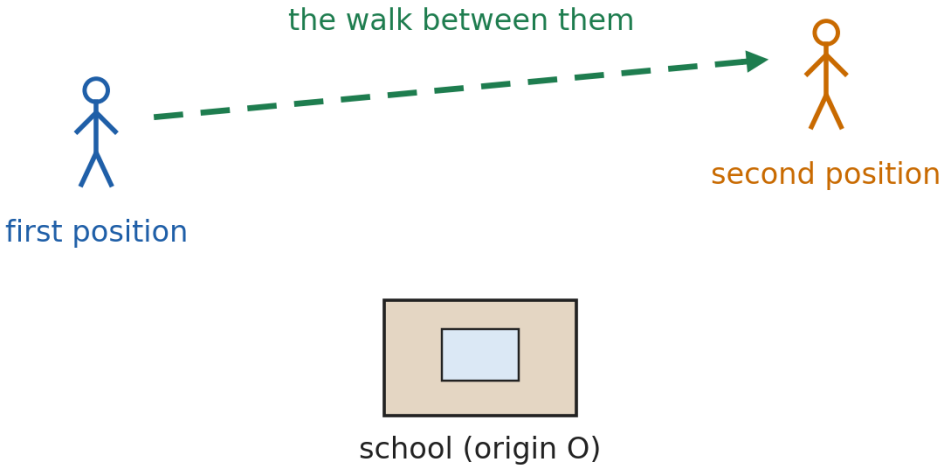
$\Delta\vec{v} = \vec{v}_f + (-\vec{v}_i)$, drawn tip to tail (it points left and slightly down)

CONCEPT 6

Student: subtract position vectors

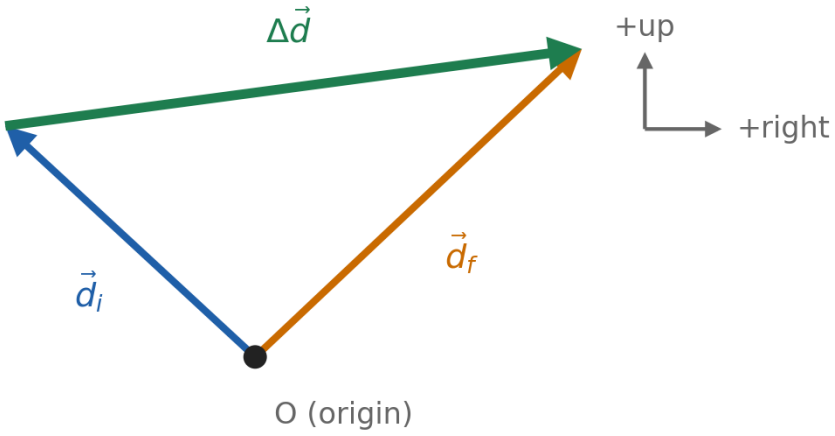
The position of a student at two points in time is shown by the vectors below. Draw a tip-to-tail diagram that would show $\vec{d}_f - \vec{d}_i$. What does this vector represent physically?

Picture the situation



A student is first at one spot near the school, then later at another spot. Both positions are measured from the school.

Positions from the origin O

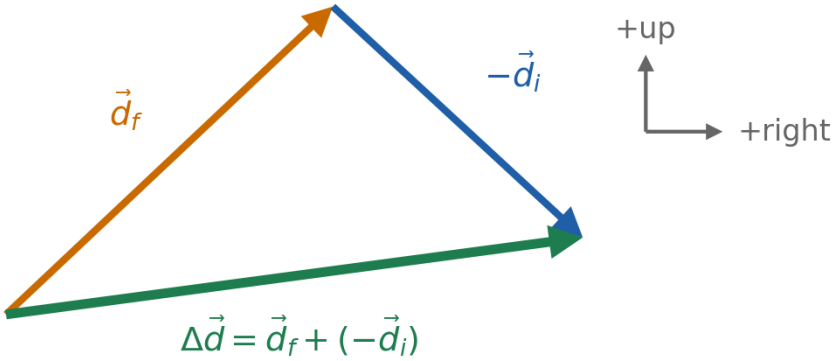


Step 1 – The tip-to-tail diagram

$\vec{d}_i$ is the student's first (initial) position and $\vec{d}_f$ is the later (final) position, both measured from O. As with velocities, rewrite the subtraction as adding the opposite. Draw $\vec{d}_f$, then from its tip draw $-\vec{d}_i$ ($\vec{d}_i$ reversed). The result goes from the first tail to the last tip.

$$\vec{d}_f - \vec{d}_i = \vec{d}_f + (-\vec{d}_i) = \Delta\vec{d}$$

Tip-to-tail: $\vec{d}_f + (-\vec{d}_i)$



Step 2 – What the arrow means

Compare with the position picture: the green arrow is the same as the arrow from the student's first position straight to the second position. That straight-line change in position is the displacement.

$\vec{d}_f - \vec{d}_i = \Delta\vec{d}$, the student's displacement (from the first position to the second)